STAT 339 METHODS OF LINEAR ALGEBRA
Vector Spaces

Edward Acheampong, PhD

(Department of Statistics & Actuarial Science)

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UNIVERSITY OF GHANA

Outline

1. Vector Spaces
   - Vectors in R^n
   - Vector Spaces
   - Subspaces of Vector Spaces
   - Spanning Sets and Linear Independence
   - Basis and Dimension
   - Rank of a Matrix and Systems of Linear Equations
   - Sum of two Subspaces

Outline

Vectors in R^n

An ordered n-tuple

An ordered n-tuple is a sequence of n real numbers $(x_1, x_2, x_3, \cdots, x_n)$.

R^n-space

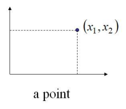
R^n-space is the set of all ordered n-tuples.

Example

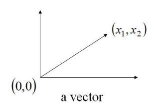
- $n = 1$: R^1-space denotes the set of all real numbers (R^1-space can be represented geometrically by the x-axis)
- $n = 2$: R^2-space denotes the set of all ordered pair of real numbers (x_1, x_2) (R^2-space can be represented geometrically by the xy-plane)
- $n = 3$: R^3-space denotes the set of all ordered pair of real numbers (x_1, x_2, x_3) (R^3-space can be represented geometrically by the xyz-plane)
- $n = 4$: R^4-space denotes the set of all ordered quadruple of real numbers (x_1, x_2, x_3, x_4)

Note

1. An n-tuple $(x_1, x_2, \cdots, x_n)$ can be viewed as a point in R^n with the x_i's as its coordinates.



2. An n-tuple $(x_1, x_2, \cdots, x_n)$ also can be viewed as a vector $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ in R^n with the x_i's as its components



A vector on the plane is expressed geometrically by a directed line segment whose initial point is the origin and whose terminal point is the point (x_1, x_2).

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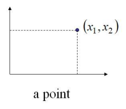

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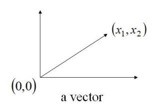

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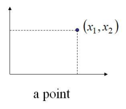

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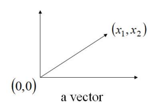

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Vectors in R^n

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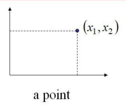

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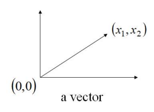

A vector on the plane is expressed geometrically by a directed line segment whose initial point is the origin and whose terminal point is the point (x_1, x_2).

Vector Operations

Let $\mathbf{u} = (u_1, u_2, \cdots, u_n)$ and $\mathbf{v} = (v_1, v_2, \cdots, v_n)$ be two vectors in R^n.

- Equality: $\mathbf{u} = \mathbf{v}$ if and only if $u_1 = v_1, u_2 = v_2, \cdots, u_n = v_n$
- Vector addition (the sum of $\mathbf{u}$ and $\mathbf{v}$): $\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \cdots, u_n + v_n)$
- Scalar multiplication (the scalar multiple of $\mathbf{u}$ by c): $c\mathbf{u} = (cu_1, cu_2, \cdots, cu_n)$
- Difference between $\mathbf{u}$ and $\mathbf{v}$: $\mathbf{u} - \mathbf{v} = (u_1 - v_1, u_2 - v_2, \cdots, u_n - v_n)$
- Zero vector: $\mathbf{0} = (0, 0, \cdots, 0)$

Note

- The sum of two vectors and the scalar multiple of a vector in R^n are called the standard operations in R^n.

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Vectors in R^n

Note

A vector $\mathbf{u} = (u_1, u_2, \cdots, u_n)$ in R^n can be viewed as:

- a $1 \times n$ row matrix (row vector):

$$\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix}$$

- a $n \times 1$ column matrix (column vector):

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

Therefore, the operations of matrix addition and scalar multiplication generate the same results as the corresponding vector operations (see the next slide)

Example (Vector addition)

- $$\begin{aligned}\mathbf{u}+\mathbf{u} &= (u_1, u_2, \cdots, u_n) + (v_1, v_2, \cdots, v_n) \\ &= (u_1 + v_1, u_2 + v_2, \cdots, u_n + v_n)\end{aligned}$$

- $$\begin{aligned}\mathbf{u}+\mathbf{u} &= \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} + \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \\ &= \begin{bmatrix} u_1 + v_1 & u_2 + v_2 & \cdots & u_n + v_n \end{bmatrix}\end{aligned}$$

- $$\mathbf{u}+\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \cdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ \cdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \cdots \\ u_n + v_n \end{bmatrix}$$

Example (Scalar multiplication)

- $$\begin{aligned} c\mathbf{u}+\mathbf{u} &= c(u_1, u_2, \cdots, u_n) \\ &= (cu_1, cu_2, \cdots, cu_n) \end{aligned}$$

- $$\begin{aligned} c\mathbf{u} &= c\begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \\ &= \begin{bmatrix} cu_1 & cu_2 & \cdots & cu_n \end{bmatrix} \end{aligned}$$

- $$c\mathbf{u} = c\begin{bmatrix} u_1 \\ u_2 \\ \cdots \\ u_n \end{bmatrix} = \begin{bmatrix} cu_1 \\ cu_2 \\ \cdots \\ cu_n \end{bmatrix}$$

Theorem ((4.2) Properties of vector addition and scalar multiplication)

Let $\mathbf{u}$, $\mathbf{v}$ *and* $\mathbf{w}$ *be a vectors in* R^n, *and let* c *and* d *be scalars, then*

1. $\mathbf{u}+\mathbf{v}$ *is a vector in* R^n *(closure under vector addition)*
2. $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$ *(commutative property of vector addition)*
3. $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{u}+\mathbf{w})$ *(associative property of vector addition)*
4. $\mathbf{u}+\mathbf{0}=\mathbf{u}$ *(additive identity property)*
5. $\mathbf{u}+(-\mathbf{u})=\mathbf{0}$ *(additive inverse property)*
6. $c\mathbf{u}$ *is a vector in* R^n *(closure under scalar multiplication)*
7. $c(\mathbf{u}+\mathbf{v})=c\mathbf{u}+c\mathbf{v}$ *(distributive property of scalar multiplication over vector addition)*
8. $(c+d)\mathbf{u}=c\mathbf{u}+d\mathbf{u}$ *(distributive property of scalar multiplication over real-number addition)*
9. $c(d\mathbf{u})=(cd)\mathbf{u}$ *(associative property of multiplication)*
10. $1\mathbf{u}=\mathbf{u}$ *(multiplicative identity property)*

Example (Practice standard vector operations in R^4)

Let $\mathbf{u} = (2, -1, 5, 0)$, $\mathbf{v} = (4, 3, 1, -1)$ and $\mathbf{w} = (-6, 2, 0, 3)$ be a vectors in R^4. Solve $\mathbf{x}$ in each of the following cases.

1. $\mathbf{x} = 2\mathbf{u} - (\mathbf{v} + 3\mathbf{w})$
2. $3(\mathbf{x} + \mathbf{w}) = 2\mathbf{u} - \mathbf{v} + \mathbf{x}$

Solution (Practice standard vector operations in R^4)

1. $\mathbf{x} = (18, -11, 9, -8)$
2. $\mathbf{x} = (9, -11/2, 9/2, -4)$

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Solution (Practice standard vector operations in R^4)

1. $\mathbf{x} = (18, -11, 9, -8)$
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Theorem ((4.3) Properties of additive identity and additive inverse)

Let $\mathbf{v}$ *be a vectors in* R^n*, and* c *be a scalars. Then the following properties are true*

1. *The additive identity is unique. That is, if* $\mathbf{u} + \mathbf{v} = \mathbf{v}$*, then* $\mathbf{u} = \mathbf{0}$.
2. *The additive inverse is unique. That is, if* $\mathbf{u} + \mathbf{v} = \mathbf{0}$*, then* $\mathbf{u} = -\mathbf{v}$.
3. $0\mathbf{v} = \mathbf{0}$
4. $c\mathbf{0} = \mathbf{0}$
5. *If* $c\mathbf{v} = \mathbf{0}$*, either* $c = 0$ *or* $\mathbf{v} = 0$
6. $-(-\mathbf{v}) = \mathbf{v}$

Note

In Properties 3 and 5 of Theorem 4.3, note that two different zeros are used, the scalar 0 and the vector $\mathbf{0}$.

Linear combination in R^n

The vector $\mathbf{x}$ is called a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, if it can be expressed in the form

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n,$$

where $c_1, c_2, \ldots, c_n$ are real numbers.

Example (Linear combination in R^3)

Given $\mathbf{x} = (-1, -2, -2)$, $\mathbf{u} = (0, 1, 4)$, $\mathbf{v} = (-1, 1, 2)$, and $\mathbf{w} = (3, 1, 2)$ in R^3, find a, b, and c such that $\mathbf{x} = a\mathbf{u} + b\mathbf{v} + c\mathbf{w}$

Solution (Linear combination in R^3)

$$\mathbf{x} = a\mathbf{u} + b\mathbf{v} + c\mathbf{w}$$

Solving yield the following equations

$$\begin{array}{ccccccc} & - & b & + & 3c & = & -1 \\ a & + & b & + & c & = & -2 \\ 4a & + & 2b & + & 2c & = & -2 \end{array}$$

Using the techniques of lectures 1,2 and 3, solve for and get

$$a = 1, b = 2, \text{ and } c = -1.$$

$\mathbf{x}$ *can be as linear combination of* $\mathbf{u}, \mathbf{v},$ *and* $\mathbf{w}$

$$\mathbf{x} = \mathbf{u} + 2\mathbf{v} - \mathbf{w}$$

Vector Spaces

Let V be a set on which two operations (vector addition and scalar multiplication) are defined. If the listed axioms are satisfied for every $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ in V and every scalar (real number) c and d, then V is called a vector space, and the elements in V are called vectors

- Addition:
  1. $\mathbf{u} + \mathbf{v}$ is in V
  2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
  3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{v} + \mathbf{v}) + \mathbf{w}$
  4. V has a zero vector $\mathbf{0}$ such that for every $\mathbf{u}$ in V, $\mathbf{u} + \mathbf{0} = \mathbf{u}$
  5. For every $\mathbf{u}$ in V, there is a vector in V denoted by $-\mathbf{u}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
- Scalar multiplication:
  1. $c\mathbf{u}$ is in V
  2. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
  3. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
  4. $c(d\mathbf{u}) = (cd)\mathbf{u}$
  5. $1(\mathbf{u}) = \mathbf{u}$

Vector Spaces

Note

A vector space consists of four entities:
a set of vectors, a set of real-number scalars, and two operations

1. V: nonempty set of vectors
2. c: any scalar
3. $+(\mathbf{u}, \mathbf{v}) = \mathbf{u} + \mathbf{v}$
4. $\cdot(c, \mathbf{u}) = c\mathbf{u}$

The set V together with the definitions of vector addition and scalar multiplication satisfying the above ten axioms is called a vector space, and it denoted by,

$$(V, +, \cdot)$$

Vector Spaces

Example (Four examples of vector spaces are shown as follows)

It is straightforward to show that these vector spaces satisfy the above ten axioms

1. n-tuple space: $V = R^n$

$$(u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n) = (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n,$$
$$k(u_1, u_2, \ldots, u_n) = (ku_1, ku_2, \ldots, ku_n)$$

2. Matrix space: $V = M_{m,n}$, the set of all $m \times n$ matrices with real-number entries

$$\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & 22 \end{bmatrix} = \begin{bmatrix} u_{11} + v_{11} & u_{12} + v_{12} \\ u_{21} + v_{21} & u_{22} + v_{22} \end{bmatrix}$$
$$k\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix}$$

Example (Four examples of vector spaces are shown as follows)

It is straightforward to show that these vector spaces satisfy the above ten axioms

3. n-th degree or less polynomial space: $V = P_n$, the set of all real-valued polynomials of degree n or less

$$\begin{aligned} p(x) + q(x) =& (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n \\ kp(x) =& ka_0 + ka_0 x + \cdots + ka_n x^n \end{aligned}$$

4. Continuous function space space: $V = C(-\infty, \infty)$, the set of all real-valued continuous functions defined on the entire real line

$$\begin{aligned} (f + g)(x) =& f(x) + g(x) \\ (kf)(x) =& f(x) \end{aligned}$$

Example (Summary of important vector spaces)

- R =set of all real numbers
- R^2 =set of all ordered pairs
- R^3 =set of all ordered triples
- R^n =set of all n-tuples
- $C(-\infty, \infty)$ =set of all continuous functions defined on the number line
- $C[a, b]$ =set of all continuous functions defined on a closed interval $[a, b]$
- P =set of all polynomials
- P_n =set of all polynomials of degree $\leq n$
- $M_{m,n}$ =set of all $m \times n$ matrices
- $M_{n,n}$ =set of all $n \times n$ square matrices

Note

To show that a set is not a vector space, you need only find one axiom that is not satisfied

Example (The set of all integers is not a vector space)

Proof.

$1 \in V$, and $\frac{1}{2}$ is a real-number scalar

$$(\frac{1}{2})(1) = \frac{1}{2} \notin V$$

(it is not closed under scalar multiplication) □

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Example (The set of all (exact) second-degree polynomial functions is not a vector space)

Proof.

Let $p(x) = x^2$, and $q(x) = -x^2 + x + 1$ is a real-number scalar

$$p(x) + q(x) = x + 1 \notin V$$

(it is not closed under vector addition) $\square$

Vector Spaces

Example

$V = R^2$ is the set of all ordered pairs of real numbers, with the standard operation of addition and the nonstandard definition of scalar multiplication listed below.

$$(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$$
$$c(u_1, u_2) = (cu_1, 0)$$

Verify V is not a vector space

Solution

This kind of setting can satisfy the first nine axioms of the definition of a vector space (you can try to show that), but it violates the tenth axiom

$$\because 1(1,1) = (1,0) \neq (1,1)$$

Therefore, the set (together with the two given operations) is not a vector space

Theorem ((4.4) Properties of additive identity and additive inverse)

Let $\mathbf{v}$ *be any element of a vector space* V*, and let* c *be any scalar. Then the following properties are true*

1. $0\mathbf{v} = \mathbf{0}$
2. $c\mathbf{0} = \mathbf{0}$
3. *if* $c\mathbf{v} = \mathbf{0}$*, either* $c = 0$ *or* $\mathbf{v} = 0$
4. $(-1)\mathbf{v} = -\mathbf{v}$ *(the additive inverse of* $\mathbf{v}$ *equals* $((-1)\mathbf{v})$

Properties of additive identity and additive inverse.

1. $$0\mathbf{v} = (c + (-c))\mathbf{v} \stackrel{(8)}{=} c\mathbf{v} + (-c)\mathbf{v} \stackrel{(9)}{=} c\mathbf{v} + (-(c\mathbf{v})) \stackrel{(5)}{=} \mathbf{0}$$

2. $$\begin{aligned} c\mathbf{0} &\stackrel{(4)}{=} c(\mathbf{0}+\mathbf{0}) \stackrel{(7)}{=} c\mathbf{0} + c\mathbf{0} \\ \implies c\mathbf{0} + (-c\mathbf{0}) &= (c\mathbf{0} + c\mathbf{0}) + (-c\mathbf{0}) \\ \stackrel{(3)}{\implies} c\mathbf{0} + (-c\mathbf{0}) &= c\mathbf{0} + [c\mathbf{0} + (-c\mathbf{0})] \\ \stackrel{(5)}{\implies} \mathbf{0} &= c\mathbf{0} + \mathbf{0} \\ \stackrel{(5)}{\implies} \mathbf{0} &= c\mathbf{0} \end{aligned}$$

Properties of additive identity and additive inverse.

3. Prove by contradiction: Suppose that $c\mathbf{v} = \mathbf{0}$, but $c \neq 0$ and $\mathbf{v} \neq \mathbf{0}$

$$\mathbf{v} \overset{(10)}{=} (1)\mathbf{v} = (\frac{1}{c}c)\mathbf{v} \overset{(9)}{=} (\frac{1}{c})(c\mathbf{v}) = (\frac{1}{c})(\mathbf{0}) = \mathbf{0} \text{ (By the second property, } c\mathbf{0} = \mathbf{0})$$

$$\implies \rightarrow\leftarrow \implies \text{ if } c\mathbf{v} = \mathbf{0}, \text{ either } c = 0 \text{ or } \mathbf{v} = \mathbf{0}$$

4.

$$0\mathbf{v} = (1 + (-1))\mathbf{v} \overset{(8)}{=} 1\mathbf{v} + (-1)\mathbf{v}$$

$$\implies \mathbf{0} = \mathbf{v} + (-1)\mathbf{v} \text{ (By the first property, } 0\mathbf{0} = \mathbf{0})$$

$$\overset{(5)}{\implies} (-1)\mathbf{v} = -\mathbf{v} \text{ (By comparing with Axiom (5), (-1)}\mathbf{v} \text{ is the additive inverse of } \mathbf{v})$$

□

Subspace :

- A vector space

$$(V, +, \cdot)$$

- A nonempty subset of V

$$\left.\begin{aligned} W &\neq \emptyset \\ W &\subseteq V \end{aligned}\right\}$$

- The nonempty subset W is called a subspace if W is a vector space under the operations of vector addition and scalar multiplication defined on V

$$(W, +, \cdot)$$

Subspaces of Vector Spaces

Trivial subspace

Every vector space V has at least two subspaces

1. Zero vector space 0 is a subspace of V
2. V is a subspace of V

Note

Any subspaces other than these two are called proper (or nontrivial) subspaces

Subspaces of Vector Spaces

Examination of whether W being a subspace

- Since the vector operations defined on W are the same as those defined on V, and most of the ten axioms inherit the properties for vector operations, it is not needed to verify those axioms
- Therefore, the following theorem tells us it is sufficient to test for the closure conditions under vector addition and scalar multiplication to identify that a nonempty subset of a vector space is a subspace

Theorem ((4.5) Test whether a nonempty subset being a subspace)

If W is a nonempty subset of a vector space V, then W is a subspace of V if and only if the following conditions hold

1. *If $\mathbf{u}$ and $\mathbf{v}$ are in W, then $\mathbf{u} + \mathbf{v}$ is in W*
2. *If $\mathbf{u}$ is in W and c is any scalar, then $c\mathbf{u}$ is in W*

(4.5) Test whether a nonempty subset being a subspace.

1. Note that if $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are in W, then they are also in V. Furthermore, W and V share the same operations. Consequently, vector space axioms 2, 3, 7, 8, 9, and 10 are satisfied automatically
2. Suppose that the closure conditions hold in Theorem 4.5, i.e., the axioms 1 and 6 for vector spaces are satisfied
3. Since the axiom 6 is satisfied (i.e., $c\mathbf{u}$ is in W if $\mathbf{u}$ is in W), we can obtain
    1. for a scalar $c = 0$, $c\mathbf{u} = \mathbf{0} \in W \implies \exists$ zero vector in $W \implies$ axiom 4 is satisfied
    2. for a scalar $c = -1$, $(-1)\mathbf{u} \in W \implies \exists -\mathbf{u} = (-1)\mathbf{u}$ such that

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{u} + (-1)\mathbf{u} = \mathbf{0}$$

$\implies$ axiom 5 is satisfied

□

Subspaces of Vector Spaces

Example (A subspace of $M_{2\times 2}$)

Let W be the set of all 2×2 symmetric matrices. Show that W is a subspace of the vector space $M_{2\times 2}$, with the standard operations of matrix addition and scalar multiplication

Solution

First, we know that W, the set of all 2×2 symmetric matrices, is an nonempty subset of the vector space $M_{2\times 2}$.
Second,

$$A_1 \in W, A_2 \in W \implies (A_1 + A_2)^T = A_1^T + A_2^T = A_1 + A_2$$
$$c \in R, A \in W \implies (cA)^T = cA^T = cA$$

Thus, Theorem 4.5 is applied to obtain that W is a subspace of $M_{2\times 2}$.

Example (A subspace of $M_{2\times 2}$)

Let W be the set of all 2×2 symmetric matrices. Show that W is a subspace of the vector space $M_{2\times 2}$, with the standard operations of matrix addition and scalar multiplication

Solution

First, we know that W, the set of all 2×2 symmetric matrices, is an nonempty subset of the vector space $M_{2\times 2}$.
Second,

$$\begin{aligned} A_1 \in W, A_2 \in W &\implies (A_1 + A_2)^T = A_1^T + A_2^T = A_1 + A_2 \\ c \in R, A \in W &\implies (cA)^T = cA^T = cA \end{aligned}$$

Thus, Theorem 4.5 is applied to obtain that W is a subspace of $M_{2\times 2}$.

Example (The set of singular matrices is not a subspace of $M_{2\times 2}$)

Let W be the set of singular (noninvertible) matrices of order 2. Show that W is not a subspace of $M_{2\times 2}$, with the standard matrix operations.

Solution

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in W, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in W$$

$$\because A + B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \notin W (W \textit{ is not closed under vector addition})$$

$\therefore$ *W is not a subspace of $M_{2\times 2}$.*

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$\therefore W$ *is not a subspace of* $M_{2\times 2}$.

Example (The set of first-quadrant vectors is not a subspace of R^2)

Show that $W = \{(x_1, x_2) : x_1 \leq 0 \text{ and } x_2 \leq 0\}$, with the standard matrix operations, is not a subspace of R^2

Solution

Let $\mathbf{u} = (1, 1) \in W$

$\because (-1)\mathbf{u} = (-1)(1, 1) = (-1, -1) \notin W$

(W is not closed under scalar multiplication)

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Example (Identify subspaces of R^2)

Which of the following two subsets is a subspace of R^2

1. The set of points on the line given by $x + 2y = 0$
2. The set of points on the line given by $x + 2y = 1$

Solution

1.

$$
\begin{aligned}
W =& \{(x, y) | x + 2y = 0\} = \{(-2t, t) | t \in R\} \textit{(the zero vector (0,0) is on this line)} \\
Let\ \mathbf{v}_1 =& (-2t_1, t_1) \in W \ and\ \mathbf{v}_2 = (-2t_2, t_2) \in W \\
\because \mathbf{v}_1 + \mathbf{v}_2 =& (-2(t_1 + t_2), t_1 + t_2) \in W \textit{(closed under vector addition)} \\
c\mathbf{v}_1 =& (-2(ct_1), ct_1) \in W \textit{(closed under scalar multiplication)}
\end{aligned}
$$

$\therefore$ *W is a subspace of R^2.*

Subspaces of Vector Spaces

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Solution

2. $$W = \{(x, y) | x + 2y = 1\} = \{(-2t, t) | t \in R\}$$ *(the zero vector (0,0) is not on this line)*

$$Consider\ \mathbf{v} = (1, 0) \in W$$

$$\because (-1)\mathbf{v} = (-1 + 0), t_1 + t_2) \notin W$$

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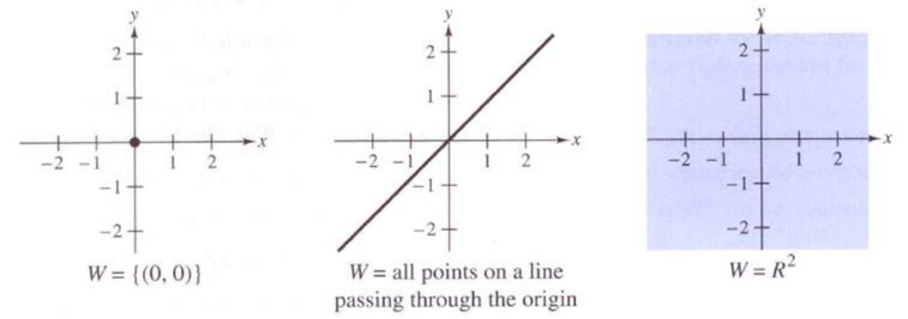
$\therefore$ *W is not a subspace of* R^2.

Subspaces of Vector Spaces

Note: Subspaces of R^2

1. W consists of the single point $\mathbf{0} = (0, 0)$. (trivial subspace)
2. W consists of alls point on a line passing through the origin.
3. R^2. (trivial subspace)



Example (Identify subspaces of R^3)

Which of the following two subsets is a subspace of R^3

1. $W = \{(x_1, x_2, 1) | x_1, x_2 \in R\}$ (Note: the zero vector is not in W)
2. $W = \{(x_1, x_1 + x_3, x_3) | x_1, x_3 \in R\}$ (Note: the zero vector is in W)

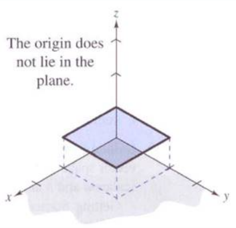
Solution

1.

$$\text{Consider } \mathbf{v} = (0, 0, 1) \in W$$

$$\because (-1)\mathbf{v} = (0, 0, -1) \notin W$$

$\therefore W$ *is not a subspace of* R^3.



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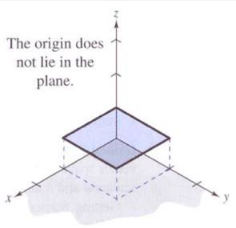
Solution

1.

$$\begin{aligned} &\textit{Consider } \mathbf{v} = (0, 0, 1) \in W \\ &\because (-1)\mathbf{v} = (0, 0, -1) \notin W \end{aligned}$$

$\therefore$ *W is not a subspace of* R^3.



Subspaces of Vector Spaces

Example (Identify subspaces of R^3)

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1. $W = \{(x_1, x_2, 1) | x_1, x_2 \in R\}$ (Note: the zero vector is not in W)
2. $W = \{(x_1, x_1 + x_3, x_3) | x_1, x_3 \in R\}$ (Note: the zero vector is in W)

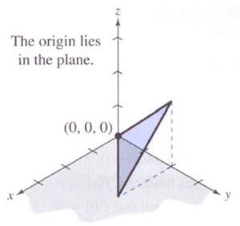
Solution

2.

$$\text{Consider } \mathbf{v} = (v_1, v_1 + v_2, v_3) \in W \text{ and } \mathbf{u} = (u_1, u_1 + u_2, u_3) \in W$$

$$\because \mathbf{v} + \mathbf{u} = (v_1 + u_1, (v_1 + v_2) + (u_1 + u_2), v_3 + u_3) \in W$$

$$c\mathbf{v} = (cv_1, (cv_1) + (cv_2), cv_3) \in W$$



$\therefore$ *W is closed under vector addition and scalar multiplication, so W is a subspace of R^3.*

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2. $W = \{(x_1, x_1 + x_3, x_3) | x_1, x_3 \in R\}$ (Note: the zero vector is in W)

Solution

2.

$$\begin{aligned} \textit{Consider } \mathbf{v} &= (v_1, v_1 + v_2, v_3) \in W \textit{ and } \mathbf{u} = (u_1, u_1 + u_2, u_3) \in W \\ \because \mathbf{v} + \mathbf{u} &= (v_1 + u_1, (v_1 + v_2) + (u_1 + u_2), v_3 + u_3) \in W \\ c\mathbf{v} &= (cv_1, (cv_1) + (cv_2), cv_3) \in W \end{aligned}$$

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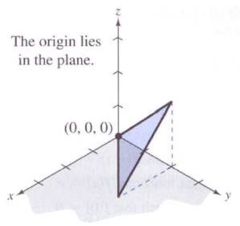

Subspaces of Vector Spaces

Note: Subspaces of R^3

1. W consists of the single point $\mathbf{0} = (0, 0, 0)$. (trivial subspace)
2. W consists of alls point on a line passing through the origin.
3. W consists of alls point on a plane passing through the origin. (The W in problem (2) is a plane passing through the origin)
4. R^3. (trivial subspace)

Theorem ((4.6) The intersection of two subspaces is a subspace)

If V and W are both subspaces of a vector space U, then the intersection of V and W (denoted by $V \cap W$) is also a subspace o U.

Proof.

1. For $\mathbf{v}_1$ and $\mathbf{v}_2$ in $V \cap W$, since $\mathbf{v}_1$ and $\mathbf{v}_2$ in V (and W), $\mathbf{v}_1 + \mathbf{v}_2$ is in V (and W). Therefore $\mathbf{v}_1 + \mathbf{v}_2$ is in $V \cap W$.
2. For $\mathbf{v}_1$ in $V \cap W$, since $\mathbf{v}_1$ is in V (and W), $c\mathbf{v}_1$ is in V (and W). Therefore $c\mathbf{v}_1$ is in $V \cap W$.

Consequently, we can conclude that the intersection of V and W ($V \cap W$ is also a subspace of U

$\square$

Spanning Sets and Linear Independence

- This section introduces the spanning set, linear independence, and linear dependence
- The above three notions are associated with the representation of any vector in a vector space as a linear combination of a selected set of vectors in that vector space

Linear combination

A vector $\mathbf{u}$ in a vector space V is called a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ in V if $\mathbf{u}$ can be written in the form

$$\mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k,$$

where $c_1, c_2, \ldots, c_k$ are real-number scalars.

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where $c_1, c_2, \ldots, c_k$ are real-number scalars.

Example (Finding a linear combination)

Given $\mathbf{v}_1 = (1, 2, 3)$, $\mathbf{v}_2 = (0, 1, 2)$ and $\mathbf{v}_3 = (-1, 0, 1)$. Prove

1. $\mathbf{w} = (1, 1, 1)$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$
2. $\mathbf{w} = (1, -2, 2)$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$

Solution (Finding a linear combination)

1.

$$\begin{aligned}
\mathbf{w} &= c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3\\
(1,1,1) &= c_1(1,2,3) + c_2(0,1,2) + c_3(-1,0,1)\\
&= (c_1 - c_3, 2c_1 + c_2, 2c_2 + c_3)
\end{aligned}$$

$$\implies \begin{array}{rcrcrcl} c_1 & & & - & c_3 & = & 1\\ 2c_1 & + & c_2 & & & = & 1\\ 3c_1 & + & 2c_2 & + & c_3 & = & 1 \end{array}$$

$$\implies \begin{bmatrix} 1 & 0 & -1 & 1\\ 2 & 1 & 0 & 1\\ 3 & 2 & 1 & 1 \end{bmatrix} \xrightarrow{G.-J.E} \begin{bmatrix} 1 & 0 & -1 & 1\\ 0 & 1 & 0 & -1\\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\implies c_1 = 1 + t, c_2 = -1 - 2t, c_3 = t$$

Solution (Finding a linear combination)

1. *This system has infinitely many solutions*

$$
\begin{aligned}
\xRightarrow{t=1}\ \mathbf{w} &= 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3 \\
\xRightarrow{t=2}\ \mathbf{w} &= 3\mathbf{v}_1 - 5\mathbf{v}_2 + 2\mathbf{v}_3 \\
&\vdots
\end{aligned}
$$

Solution (Finding a linear combination)

2.

$$\begin{aligned}
\mathbf{w} &= c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 \\
(1,-2,2) &= c_1(1,2,3) + c_2(0,1,2) + c_3(-1,0,1) \\
\implies & \begin{bmatrix} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{bmatrix} \xrightarrow{G.E} \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 7 \end{bmatrix} \\
\implies & \textit{ This system has no solution since the third row means} \\
& 0 \cdot c_1 + 0 \cdot c_2 + 0 \cdot c_3 = 7
\end{aligned}$$

$\implies$ $\mathbf{w}$ *can not be expressed as* $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$

The span of a set: span(S)

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a set of vectors in a vector space V, then the span of S is the set of all linear combinations of the vectors in S,

$$\text{span}(S) = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k | \forall c_i \in S\},$$

(the set of all linear combinations of vectors in S).

Definition of a spanning set of a vector space

If every vector in a given vector space V can be written as a linear combination of vectors in a set S, then S is called a spanning set of the vector space V

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(the set of all linear combinations of vectors in S).

Definition of a spanning set of a vector space

If every vector in a given vector space V can be written as a linear combination of vectors in a set S, then S is called a spanning set of the vector space V

Note: The above statement can be expressed as follows

$$
\begin{aligned}
& \operatorname{span}(S) = V \\
\iff & S \text{ span (generates) } V \\
\iff & V \text{ spanned (generated) by } S \\
\iff & V \text{ is spanning set of } V
\end{aligned}
$$

Example

1. The set $S = \{(1,0,0),(0,1,0),(0,0,1)\}$ spans R^3 because any vector $\mathbf{u} = (u_1, u_2, u_3)$ in R^3 can be written as

$$\mathbf{u} = u_1(1,0,0) + u_2(0,1,0) + u_3(0,0,1)$$

2. The set $S = \{1, x, x^2\}$ spans P_2 because any polynomial function $p(x) = a + bx + cx^2$ in P_2 can be written as

$$p(x) = a(1) + b(x) + c(x^2)$$

Example (A spanning set for R^3)

Show that the set $S = \{(1,2,3),(0,1,2),(-2,0,1)\}$ spans R^3.

Solution (A spanning set for R^3)

We must examine whether any vector $\mathbf{u} = (u_1, u_2, u_3)$ *in* R^3 *can be expressed as a linear combination of* $\mathbf{v}_1 = (1,2,3)$, $\mathbf{v}_2 = (0,1,2)$ *and* $\mathbf{v}_3 = (-2,0,1)$.

If $\mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$

$$\Longrightarrow \begin{array}{rcrcrcl} c_1 & & & - & 2c_3 & = & u_1 \\ 2c_1 & + & c_2 & & & = & u_2 \\ 3c_1 & + & 2c_2 & + & c_3 & = & u_3 \end{array}$$

The above problem thus reduces to determine whether this system is consistent for all values of $u_1, u_2,$ *and* u_3.

Solution (A spanning set for R^3)

$$\because |A| = \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} \neq 0$$

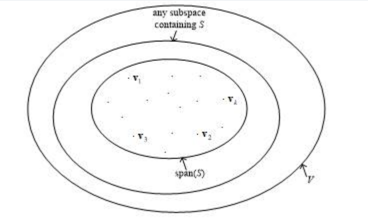
$$\therefore A\mathbf{x} = \mathbf{u} \text{ has exactly one solution for every } \mathbf{u}$$

$$\implies span(S) = R^3$$

Theorem ((4.7) span(S) is a subspace of V)

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a set of vectors in a vector space V, then

1. *span(S) is a subspace of V*
2. *span(S) is the smallest subspace of V that contains S, i.e., every other subspace of V containing S must contain span(S)*



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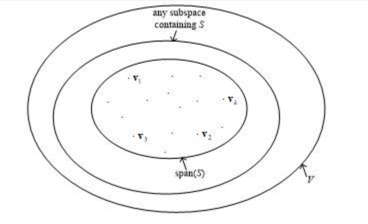

Proof.

1. Consider any two vectors $\mathbf{u}$ and $\mathbf{v}$ in span(S), that is,

$$\mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k \text{ and } \mathbf{v} = d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \cdots + d_k\mathbf{v}_k$$

Then $\mathbf{u} + \mathbf{v} = (c_1 + d_1)\mathbf{v}_1 + (c_2 + d_2)\mathbf{v}_2 + \cdots + (c_k + d_k)\mathbf{v}_k \in \text{span}(S)$ and $c\mathbf{u} = (cc_1)\mathbf{v}_1 + (cc_2c)\mathbf{v}_2 + \cdots + (cc_kc)\mathbf{v}_k \in \text{span}(S)$ because they can be written as linear combinations of vectors in S

So, according to Thm 4.5, we can conclude that span(S) is a subspace of V.

$\square$

Proof.

2. Let U be another subspace of V that contains S, and we want to show that $\text{span}(S) \subset U$ Consider any $\mathbf{u} \in \text{span}(S)$, that

$$\mathbf{u} = \sum_{i=1}^{k} c_i \mathbf{v}_i$$

where $\mathbf{v}_i \in S$ U contain $S \implies \mathbf{v}_i \in U \overset{U \text{ is a subspace}}{\Longrightarrow} \mathbf{u} = \sum_{i=1}^{k} c_i \mathbf{v}_i \in U$
Since for any vector $\mathbf{u} \in \text{span}(S)$, $\mathbf{u}$ also belongs to U, we conclude that $\text{span}(S) \subset U$, and therefore $\text{span}(S)$ are the smallest subspace of V that contains $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$.

$\square$

Definitions of Linear Independence (L.I.) and Linear Dependence (L.D.)

$S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$; a set of vectors in a vector space V

For $\sum_{i=1}^{k} c_i \mathbf{v}_i = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = 0$

1. If the equation has only the trivial solution $(c_1 = c_2 = \cdots = c_k = 0)$, then S (or $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$) is called **linearly independent**
2. If the equation has a nontrivial solution (that is, not all c's are zero), then S (or $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$) is called **linearly dependent** (The name of linear dependence is from the fact that in this case, there exist at least one $\mathbf{v}_i$ which can be represented by the linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ in which the coefficients are not zeros.)

Example (Testing for linear independence)

Determine whether the following set of vectors in R^3 is L.I. or L.D

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1,2,3), (0,1,2), (-2,0,1)\}$$

Solution

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$$

$$\implies \begin{array}{rcrcrcl} c_1 & & & - & 2c_3 & = & 0 \\ 2c_1 & + & c_2 & & & = & 0 \\ 3c_1 & + & 2c_2 & + & c_3 & = & 0 \end{array}$$

$$\implies \begin{bmatrix} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \xleftrightarrow{G.-J.E.} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

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$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1,2,3), (0,1,2), (-2,0,1)\}$$

Solution

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$$

$$\Longrightarrow \begin{array}{rcrcrcl} c_1 & & & - & 2c_3 & = & 0 \\ 2c_1 & + & c_2 & & & = & 0 \\ 3c_1 & + & 2c_2 & + & c_3 & = & 0 \end{array}$$

$$\Longrightarrow \begin{bmatrix} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \xleftrightarrow{G.-J.E.} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Solution

$$\implies c_1 = c_2 = c_3 = 0 \textit{ (only the trivial solution)}$$

$$\textit{(or det(A)=-1}\neq 0\textit{, so there is only the trivial solution)}$$

$$\implies S(\textit{ or } \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) \textit{ are linearly independent}$$

Example (Testing for linear independence)

Determine whether the following set of vectors in P_2 is L.I. or L.D

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{1 + x - 2x^2, 2 + 5x - x^2, x + x^2\}$$

Solution

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$$

$$c_1(1 + x - 2x^2) + c_2(2 + 5x - x^3) + c_3(x + x^2) = 0 + 0x - 0x^2$$

$$\Longrightarrow \begin{array}{rcrcrcl} c_1 & + & 2c_2 & & & = & 0 \\ c_1 & + & 5c_2 & + & c_3 & = & 0 \\ -2c_1 & - & c_2 & + & c_3 & = & 0 \end{array} \Longrightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{bmatrix} \overset{G.E.}{\longleftrightarrow} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 1/3 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Example (Testing for linear independence)

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$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{1 + x - 2x^2, 2 + 5x - x^2, x + x^2\}$$

Solution

$$\begin{aligned} c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 &= 0 \\ c_1(1 + x - 2x^2) + c_2(2 + 5x - x^3) + c_3(x + x^2) &= 0 + 0x - 0x^2 \end{aligned}$$

$$\implies \begin{array}{rcrcrcl} c_1 & + & 2c_2 & & & = & 0 \\ c_1 & + & 5c_2 & + & c_3 & = & 0 \\ -2c_1 & - & c_2 & + & c_3 & = & 0 \end{array} \implies \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{bmatrix} \xleftrightarrow{G.E.} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 1/3 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Solution

$\implies$ *This system has infinitely many solutions (i.e., this system has nontrivial solutions, e.g., $c_1 = 2, c_2 = -1, c_3 = 3$)*

$\implies$ *S (or $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$) are linearly dependent*

Example (Testing for linear independence)

Determine whether the following set of vectors in 2×2 is L.I. or L.D

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right\}$$

Solution

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$$

$$c_1 \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Longrightarrow \begin{array}{rcrcrcl} c_1 & + & 2c_2 & & & = & 0 \\ c_1 & + & 5c_2 & + & c_3 & = & 0 \\ -2c_1 & - & c_2 & + & c_3 & = & 0 \end{array} \Longrightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{bmatrix} \xleftrightarrow{G.E.} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 1/3 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Example (Testing for linear independence)

Determine whether the following set of vectors in 2×2 is L.I. or L.D

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right\}$$

Solution

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$$

$$c_1 \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Longrightarrow \begin{array}{rcrcrcl} c_1 & + & 2c_2 & & & = & 0 \\ c_1 & + & 5c_2 & + & c_3 & = & 0 \\ -2c_1 & - & c_2 & + & c_3 & = & 0 \end{array} \Longrightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{bmatrix} \xleftrightarrow{G.E.} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 1/3 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Solution

$$\Longrightarrow \begin{array}{ccccccc} 2c_1 & + & 3c_2 & + & c_3 & = & 0 \\ c_1 & & & & & = & 0 \\ & & 2c_2 & + & 2c_3 & = & 0 \\ c_1 & + & c_2 & & & = & 0 \end{array} \Longrightarrow \begin{bmatrix} 2 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \overset{G.-J.E.}{\longleftrightarrow} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution

$\Longrightarrow$ *This system has only the trivial solution (i.e., this system has nontrivial solutions, e.g.,* $c_1 = 0, c_2 = 0, c_3 = 0$*)*

$\Longrightarrow$ *S (or $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$) are linearly independent*

Theorem ((4.8) A property of linearly dependent sets)

A set $S == \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$, for $k \geq 2$, is linearly dependent if and only if at least one of the vectors $\mathbf{v}_i$ in S can be written as a linear combination of the other vectors in S

Proof.

$(\Rightarrow)$

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = 0$$

$$\because S \text{ is linearly dependent (there exist nontrivial solutions)}$$

$$\implies c_i \neq 0 \text{ for some } i$$

$$\implies \mathbf{v}_i = (-\frac{c_1}{c_i})\mathbf{v}_1 + \cdots + (-\frac{c_{i-1}}{c_i})\mathbf{v}_{i-1} + (-\frac{c_{i+1}}{c_i})\mathbf{v}_{i+1} + \cdots + (-\frac{c_k}{c_i})\mathbf{v}_k$$

$\square$

Proof.

$(\Leftarrow)$

$$\begin{aligned}
\text{Let } \mathbf{v}_i &= d_1\mathbf{v}_1 + \cdots + d_{i-1}\mathbf{v}_{i-1} + d_{i+1}\mathbf{v}_{i+1} + \cdots + d_k\mathbf{v}_k \\
d_1\mathbf{v}_1 + \cdots + d_{i-1}\mathbf{v}_{i-1} + d_{i+1}\mathbf{v}_{i+1} + \cdots + d_k\mathbf{v}_k &= \mathbf{0} \\
\implies c_1 = d_1, c_2 = d_2, \ldots, c_i = -1, \ldots, c_k = d_k &\text{(there exits at least this nontrivial solution)} \\
&\implies S \text{ is linearly dependent}
\end{aligned}$$

□

Corollary to Theorem 4.8

(A corollary is a must-be-true result based on the already proved theorem)

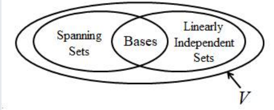
Corollary to Theorem 4.8

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in a vector space V are linearly dependent (for $k = 2$ in Theorem 4.8) if and only if one is a scalar multiple of the other

Basis and Dimension

Theorem (Basis)

- *V: a vector space*
- $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} \subseteq V$
  1. *S spans V (i.e., $span(S) = V$)*
     (For any $\mathbf{u} \in V, \sum c_i\mathbf{v}_i = A\mathbf{x} = \mathbf{u}$ has at least one solution
     (If there is exact one solution ($det(A) \neq 0$)
     or if there are infinitely many solution ($det(A) = 0$)))
  2. *S is linearly independent*
     (For $\sum c_i\mathbf{v}_i = A\mathbf{x} = \mathbf{0}$, there is only the trivial solution ($det(A) \neq 0$)



$\therefore$ S is called a basis for V (i.e., $\sum c_i\mathbf{v}_i = A\mathbf{x}$, ($det(A) \neq 0$)

Corollary to Theorem 4.8

A basis S must have enough vectors to span V, but not so many vectors that one of them could be written as a linear combination of the other vectors in S

Example (Basis)

1. the standard basis for R^2
   $\{i, j\}$, for $i = (1, 0), j = (0, 1)$
2. the standard basis for R^3
   $\{i, j, k\}$, for $i = (1, 0, 0), j = (0, 1, 0), k = (0, 0, 1)$
3. the standard basis for R^n
   $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$, for $\mathbf{e}_1 = (1, 0, \ldots, 0), \mathbf{e}_2 = (0, 1, \ldots, 0), \ldots, \mathbf{e}_n = (0, 0, \ldots, 1)$

Note

Express any vector in R^n as the linear combination of the vectors in the standard basis: the coefficient for each vector in the standard basis is the value of the corresponding component of the examined vector, e.g., (1, 3, 2) can be expressed as $1{\cdot}(1, 0, 0) + 3{\cdot}(0, 1, 0) + 2{\cdot}(0, 0, 1)$

Basis and Dimension

Example (Basis)

4. the standard basis for $m \times n$ matrix space

   $\{E_{ij} | 1 \leq i \leq m, 1 \leq j \leq n\}$, and $E_{ij} = \begin{cases} a_{ij} = 1 \\ \text{other entries are zero} \end{cases}$

   Eg. $m \times n$ matrix space

   $$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

5. the standard basis for $P_n(x)$

   $\{1, x, x^2, \ldots, x^n\}$

   E.g.

   $P_3(x) : \{1, x, x^2, x^3\}$

Example (The nonstandard basis for R^2)

Show that $S = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(1,1),(1,-1)\}$ is a basis for R^2.

1. For any $\mathbf{u} = (u_1, u_2) \in R^2$, $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{u} \implies = \begin{cases} c_1 + c_2 = u_1 \\ c_1 - c_2 = u_2 \end{cases}$

   Because the coefficient matrix of this system has a nonzero determinant, the system has a unique solution for each $\mathbf{u}$. Thus you can conclude that S spans R^2

2. For $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0} \implies = \begin{cases} c_1 + c_2 = 0 \\ c_1 - c_2 = 0 \end{cases}$

   Because the coefficient matrix of this system has a nonzero determinant, you know that the system has only the trivial solution. Thus you can conclude that S is linearly independent

According to the above two arguments, we can conclude that S is a (nonstandard) basis for R^2

Basis and Dimension

Theorem ((4.9) Uniqueness of basis representation for any vectors)

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for a vector space V, then every vector in V can be written in one and only one way as a linear combination of vectors in S

Proof.

$$\because S \text{ is a basis} \implies \begin{cases} (1)\text{span}(S) = V \\ (2)\text{S is linearly independent} \end{cases}$$

$\because \text{span}(S) = V$, Let

$$\begin{array}{ccccccccc} \mathbf{v} & = & c_1\mathbf{v}_1 & + & c_2\mathbf{v}_2 & + & \cdots & + & c_n\mathbf{v}_n \\ \mathbf{v} & = & b_1\mathbf{v}_1 & + & b_2\mathbf{v}_2 & + & \cdots & + & b_n\mathbf{v}_n \end{array}$$

$$\implies \mathbf{v} + (-1)\mathbf{v} = \mathbf{0} = (c_1 - b_1)\mathbf{v}_1 + (c_2 - b_2)\mathbf{v}_2 + \cdots + (c_n - b_n)\mathbf{v}_n$$

$\because S$ is linearly independent $\implies$ with only the trivial solution

$\implies$ coefficients for $\mathbf{v}_i$ are all zeros

Basis and Dimension

Proof.

$$\implies c_1 = b_1, c_2 = b_2, \ldots, c_n = b_n \text{ (i.e., unique basis representation)}$$

$\square$

Basis and Dimension

Theorem ((4.10) Bases and linear dependence)

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for a vector space V, then every set containing more than n vectors is linearly dependent (In other words, every linearly independent set contains at most n vectors

Proof.

Let $S_1 = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}, m > n$

$$\because \text{span}(S) = V, \text{ for } \mathbf{u}_i \in V$$

$$\begin{array}{ccccccccc}
\mathbf{u}_1 & = & c_{11}\mathbf{v}_1 & + & c_{21}\mathbf{v}_2 & + & \cdots & + & c_{n1}\mathbf{v}_n \\
\mathbf{u}_2 & = & c_{12}\mathbf{v}_1 & + & c_{22}\mathbf{v}_2 & + & \cdots & + & c_{n2}\mathbf{v}_n \\
 & & & & \vdots & & & & \\
\mathbf{u}_m & = & c_{1m}\mathbf{v}_1 & + & c_{2m}\mathbf{v}_2 & + & \cdots & + & c_{nm}\mathbf{v}_n
\end{array}$$

Proof.

Consider $k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \cdots + k_m\mathbf{u}_m = \mathbf{0}$ (if k_i's are not all zero, S_1 is linearly dependent)

$$\implies d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \cdots + d_n\mathbf{v}_n = \mathbf{0}(d_{i1} = c_{i1}k_1 + c_{i2}k_2 + \cdots + c_{im}k_m)$$

$$\because S \text{ is LIN } \implies d_i = 0 \ \forall i$$

$$\begin{array}{ccccccccc} c_{11}k_1 & + & c_{12}k_2 & + & \cdots & + & c_{1m}k_m & = & 0 \\ c_{21}k_1 & + & c_{22}k_2 & + & \cdots & + & c_{2m}k_m & = & 0 \\ & & & & \vdots & & & & \\ c_{1n}k_1 & + & c_{n2}k_2 & + & \cdots & + & c_{nm}k_n & = & 0 \end{array}$$

$\because$ Theorem 1.1: If the homogeneous system has fewer equations (n equations) than variables $(k_1, k_2, \ldots, k_m)$, then it must have infinitely many solutions

$\therefore m > n \implies k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \cdots + k_m\mathbf{u}_m = 0$ has nontrivial (nonzero) solution

$\implies S_1$ is linearly dependent

Basis and Dimension

Theorem

(4.11) Number of vectors in a basis If a vector space V has one basis with n vectors, then every basis for V has n vectors

Proof.

According to Thm. 4.10, every linearly independent set contains at most n vectors in a vector space if there is a basis of n vectors spanning that vector space. Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \text{ and } S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$$

are two bases with different sizes for V

$$\left.\begin{array}{l} \left.\begin{array}{l} S \text{ is a basis spanning } V \\ S' \text{ is a set of LIN vectors} \end{array}\right\} \implies m \leq n \\ \left.\begin{array}{l} S' \text{ is a basis spanning } V \\ S \text{ is a set of LIN vectors} \end{array}\right\} \implies n \leq m \end{array}\right\} \implies n = m$$

$\square$

Basis and Dimension

Dimension

The dimension of a vector space V is defined to be the number of vectors in a basis for V

- V: a vector space
- S: a basis for V
- $\Longrightarrow \quad \dim(V) = \#(S)$ (the number of vectors in a basis S)

Finite dimensional

A vector space V is finite dimensional if it has a basis consisting of a finite number of elements

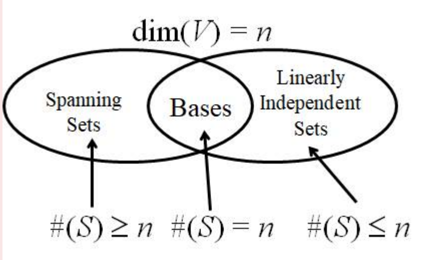
Infinite dimensional

If a vector space V is not finite dimensional, then it is called infinite dimensional

Basis and Dimension

NOTE

1. $\dim(\{\mathbf{0}\}) = 0$
2. Given $\dim(V) = n$, for $S \subseteq V$
   - S: a spanning set $\implies \#(S) \geq n$
   - S: a LIN set $\implies \#(S) \leq n$
   - S: a basis $\implies \#(S) = n$
3. Given $\dim(V) = n$, if W is a subspace of $V \implies \dim(W) \leq n$



Example (Find the dimension of a vector space according to the standard basis)

The simplest way to find the dimension of a vector space is to count the number of vectors in the "standard" basis for that vector space

1.
$$\begin{aligned} \text{Vector space } R^n &\implies \text{standard basis } \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\} \\ &\implies \dim(R^n) = n \end{aligned}$$

2.
$$\begin{aligned} \text{Vector space } P_n(x) &\implies \text{standard basis } \{1, x, x^2, \ldots, x^n\} \\ &\implies \dim(P_n(x)) = n + 1 \end{aligned}$$

Example (Find the dimension of a vector space according to the standard basis)

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2.

$$\begin{aligned}\text{Vector space } P_n(x) &\implies \text{standard basis } \{1, x, x^2, \ldots, x^n\} \\ &\implies \dim(P_n(x)) = n + 1\end{aligned}$$

Example (Find the dimension of a vector space according to the standard basis)

3.

$$\begin{aligned}\text{Vector space } P(x) &\implies \text{standard basis } \{1, x, x^2, \dots\} \\ &\implies \dim(P(x)) = \infty\end{aligned}$$

4.

$$\begin{aligned}\text{Vector space } M_{m\times n} &\implies \text{standard basis } \{E_{ij} | i \leq i \leq m, i \leq j \leq n\} \\ \text{and in } E_{ij} &\begin{cases} a_{ij} = 1 \\ \text{other entries are zero} \end{cases} \\ &\implies \dim(M_{m\times n}) = mn\end{aligned}$$

Example (Determining the dimension of a subspace of R^3)

1. $W = \{(d, c - d, c) : c \text{ and } d \text{ are real numbers}\}$
2. $W = \{(2b, b, 0) : b \text{ is real number }\}$

Solution

(Hint: find a set of LIN vectors that spans the subspace, i.e., find a basis for the subspace.)

1.

$$\begin{aligned}(d, c-d, c) &= c(0,1,1) + d(1,-1,0) \\ \implies S &= \{(0,1,1),(1,-1,0)\}(\ S \text{ is LIN and } S \text{ spans } W)\end{aligned}$$

$$\implies \quad S \text{ is a basis for } W$$

$$\implies \quad dim(W) = \#(S) = 2$$

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Basis and Dimension

Example (Determining the dimension of a subspace of R^3)

1. $W = \{(d, c-d, c) : c \text{ and } d \text{ are real numbers}\}$
2. $W = \{(2b, b, 0) : b \text{ is real number } \}$

Solution

(Hint: find a set of LIN vectors that spans the subspace, i.e., find a basis for the subspace.)

1.

$$(d, c-d, c) = c(0,1,1) + d(1,-1,0)$$

$$\implies S = \{(0,1,1),(1,-1,0)\} (\ S \text{ is LIN and } S \text{ spans } W)$$

$$\implies S \text{ is a basis for } W$$

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Example (Determining the dimension of a subspace of R^3)

1. $W = \{(d, c-d, c) : c \text{ and } d \text{ are real numbers}\}$
2. $W = \{(2b, b, 0) : b \text{ is real number }\}$

Solution

(Hint: find a set of LIN vectors that spans the subspace, i.e., find a basis for the subspace.)

1.

$$\begin{aligned}(d, c-d, c) =& c(0,1,1) + d(1,-1,0)\\ \implies S =& \{(0,1,1),(1,-1,0)\}(\ S \text{ is LIN and } S \text{ spans } W)\end{aligned}$$

$$\implies \quad S \text{ is a basis for } W$$

$$\implies \dim(W) = \#(S) = 2$$

Basis and Dimension

Solution

2.

$$
\begin{aligned}
(2b, b, 0) =& b(2,1,0)) \\
\implies S =& \{(2,1,0)\}(\ S \text{ is LIN and } S \text{ spans } W) \\
\implies & S \text{ is a basis for } W \\
\implies dim(W) =& \#(S) = 1
\end{aligned}
$$

Example (Finding the dimension of a subspace of $M_{2\times 2}$)

Let W be the subspace of all symmetric matrices in $M_{2\times 2}$. What is the dimension of W?

Solution

$$W = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \middle| a, b, c \in R \right\}$$

$$\because \begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\implies S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} spans\ W\ and\ S\ is\ LIN$$

$$\implies S\ is\ a\ basis\ for\ W \implies textdim(W) = \#(S) = 3$$

Example (Finding the dimension of a subspace of $M_{2\times 2}$)

Let W be the subspace of all symmetric matrices in $M_{2\times 2}$. What is the dimension of W?

Solution

$$W = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \middle| a, b, c \in R \right\}$$

$$\because \begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\implies S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} spans\ W\ and\ S\ is\ LIN$$

$$\implies S\ is\ a\ basis\ for\ W \implies textdim(W) = \#(S) = 3$$

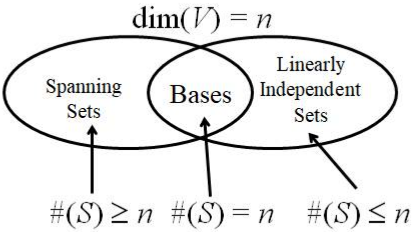
Basis and Dimension

Theorem ((4.12) Methods to identify a basis in an $n-$dimensional space)

Let V be a vector space of dimension n

1. *If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots \mathbf{v}_n\}$ is a linearly independent set of vectors in V, then S is a basis for V.*
2. *If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots \mathbf{v}_n\}$ spans V, then S is a basis for V.*

Both results are due to the fact that $\#(S) = n$



Rank of a Matrix and Systems of Linear Equations

- In this section, three vector spaces are investigated
  - Row space: the vector space spanned by the row vectors of a matrix A
  - Column space: the vector space spanned by column vectors of a matrix A
  - Nullspace: the vector space consisting of all solutions of the homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$
- Next, discuss the basis and the dimension of each vector space
- Finally, the relationship between the solutions of $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{x} = \mathbf{0}$ will be discussed
- To begin the introduction, I present the notation for the row and column vectors of a matrix A with the size $m \times n$ on the next slide

Rank of a Matrix and Systems of Linear Equations

- row vectors: (with size $1 \times n$)

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} A_{(1)} \\ A_{(2)} \\ \vdots \\ A_{(m)} \end{bmatrix}$$

$$\begin{aligned} (a_{11}\ a_{12}\ \cdots\ a_{1n}) &= A_{(1)} \\ (a_{11}\ a_{12}\ \cdots\ a_{1n}) &= A_{(1)} \\ \vdots\qquad &= \\ (a_{m1}\ a_{m2}\ \cdots\ a_{mn}) &= A_{(m)} \end{aligned}$$

- column vectors: (with size $m \times 1$)

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} A^{(1)} & A^{(2)} & \cdots & A^{(m)} \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}}_{A^{(1)}} \underbrace{\begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}}_{A^{(2)}} \cdots \underbrace{\begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}}_{A^{(n)}}$$

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Let A be an $m \times n$ matrix

- Row space of A is a subspace of R^n spanned by the row vectors of A

$$RS(A) = \{\alpha_1 A_{(1)} + \alpha_2 A_{(2)} + \cdots + \alpha_m A_{(m)} | \alpha_1, \alpha_2, \ldots, \alpha_m \in R\}$$

(If $A_{(1)}, A_{(2)}, \ldots, A_{(m)}$ are linearly independent, $A_{(1)}, A_{(2)}, \ldots, A_{(m)}$ can form a basis for $RS(A)$)

- Column space of A is a subspace of R^m spanned by the column vectors of A

$$CS(A) = \{\beta_1 A_{(1)} + \beta_2 A_{(2)} + \cdots + \beta_n A_{(m)} | \beta_1, \beta_2, \ldots, \beta_n \in R\}$$

(If $A^{(1)}, A_{(2)}, \ldots, A^{(n)}$ are linearly independent, $A^{(1)}, A^{(2)}, \ldots, A^{(n)}$ can form a basis for $CS(A)$)

Note

1. The definitions of $RS(A)$ and $CS(A)$ satisfy automatically the closure conditions of vector addition and scalar multiplication
2. $\dim(RS(A))$ or $\dim(CS(A))$ equals the number of linearly independent row (or column) vectors of A

Rank of a Matrix and Systems of Linear Equations

Theorem ((4.13) Row-equivalent matrices have the same row space)

If an $m \times n$ matrix A is row equivalent to an $m \times n$ matrix B, then the row space of A is equal to the row space of B

Proof.

1. Since B can be obtained from A by elementary row operations, the row vectors of B can be expressed as linear combinations of the row vectors of A $\implies$ The linear combinations of row vectors in B must be linear combinations of row vectors in A $\implies$ any vector in $RS(B)$ lies in $RS(A)$ $\implies$ $RS(B) \subseteq RS(A)$
2. Since A can be obtained from B by elementary row operations, the row vectors of A can be written as linear combinations of the row vectors of B $\implies$ The linear combinations of row vectors in A must be linear combinations of row vectors in B $\implies$ any vector in $RS(A)$ lies in $RS(B)$ $\implies$ $RS(A) \subseteq RS(B)$

$\therefore RA(A) = RS(B)$ $\square$

Note

1. The row space of a matrix is not changed by elementary row operations

$$RS(r(A)) = RS(A), r\text{: any elementary row operation}$$

2. But elementary row operations will change the column space

Rank of a Matrix and Systems of Linear Equations

Theorem ((4.14) Basis for the row space of a matrix)

If a matrix A is row equivalent to a matrix B in the (reduced) row-echelon form, then the nonzero row vectors of B form a basis for the row space of A

Note

1. The row space of A is the same of the row space of B (Thm. 4.13), spanned by all row vectors in B
2. For the row space of B, it can be constructed by the linear combinations of only nonzero row vectors since it is impossible to generate more combinations when taking zero row vectors into consideration (i.e., nonzero row vectors span the row space of B)
3. Since it is impossible to express a nonzero row vector as the linear combination of other nonzero row vectors in a row-echelon form matrix, according to Thm. 4.8, we can conclude that the nonzero row vectors in B are linearly independent

Theorem ((4.14) Basis for the row space of a matrix)

If a matrix A is row equivalent to a matrix B in the (reduced) row-echelon form, then the nonzero row vectors of B form a basis for the row space of A

Note

4. As a consequence, since the nonzero row vectors in B are linearly independent and span the row space of B, according to the definition on Slide 67, they form a basis for the row space of B and for the row space of A as well

Example (Finding a basis for a row space)

Find a basis of the row space of

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & 2 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & 2 \end{bmatrix} \xrightarrow{G.E.} B = \begin{array}{c} \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \mathbf{w}_3 \\ \\ \\ \end{array}$$

A basis for $RS(A) = \{$ *the nonzero row vectors of* $B\}$ *(Thm 4.14)*

$= \mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3 = \{(1, 3, 1, 3), (0, 1, 1, 0), (0, 0, 0, 1)\}$

(Check: $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ *are linearly independent, i.e.,* $a\mathbf{w}_1 + b\mathbf{w}_2 + c\mathbf{w}_3 = \mathbf{0}$ *has only the trivial solution or it is impossible to express any one of them to be the linear combination of the others (Theorem 4.8))*

Note

Although row operations can change the column space of a matrix (mentioned in Slides 89, 90 % 91), they do not change the dependency relationships among columns

1. $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_4\}$ is LIN (because these columns have the leading 1's)
   $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4\}$ is LIN
2. $\mathbf{b}_3 = -2\mathbf{b}_1 + \mathbf{b}_1 \implies \mathbf{a}_3 = -2\mathbf{a}_1 + \mathbf{a}_2$
   (The linear combination relationships among column vectors in B still hold for column vectors in A)

Example (Finding a basis for a subspace using Thm. 4.14)

Find a basis for the subspace of R^3 spanned by

$$S = \{(-1, 2, 5), (3, 0, 3), (5, 1, 8)\}$$

Solution

$$A = \begin{array}{c} \begin{bmatrix} -1 & 2 & 5 \\ 3 & 0 & 3 \\ 5 & 1 & 8 \end{bmatrix} \begin{array}{l} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \end{array} \end{array} \xrightarrow{G.E.} B = \begin{bmatrix} 1 & -2 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \end{array}$$

A basis for span $(\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\})$

$=$ *a basis for* $RS(A)$

$= \{$ *the nonzero row vectors of* $B\}$ *(Thm 4.14)*

$= \{\mathbf{w}_1, \mathbf{w}_2\}$

$= \{(1, -2, -5), (0, 1, 3)\}$

Example (Finding a basis for a row space)

Find a basis of the row space of

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & 2 \end{bmatrix}$$

Solution

$\because$ *Since* $CS(A) = RS(A^T)$, *to find a basis for the column space of the matrix* A *is equivalent to find a basis for the row space of the matrix* A^T

$$A^T = \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 3 & 1 & 0 & 4 & 0 \\ 1 & 1 & 6 & -2 & -4 \\ 3 & 0 & -1 & 1 & -2 \end{bmatrix} \xrightarrow{G.E.} B = \begin{array}{c} \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 0 & 1 & 9 & -5 & -6 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \mathbf{w}_3 \\ \\ \\ \end{array}$$

$\therefore$ *a basis for* $CS(A)$
$=$ *a basis for* $RS(A^T)$
$= \{$ *the nonzero row vectors of* $B\}$ *(Thm 4.14)*

Solution

$$= \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$$

$$= \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 9 \\ -5 \\ -6 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ -1 \end{bmatrix} \right\} \textit{(a basis for the column space of } A\textit{)}$$

Rank of a Matrix and Systems of Linear Equations

Theorem ((4.16) The definition of the nullspace)

If A is $m \times n$ matrix, then the set of all solutions of the homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$ is a subspace of R^n called the nullspace of A, which is denoted as

$$NS(A) = \{\mathbf{x} \in R^n | A\mathbf{x} = \mathbf{0}\}$$

Proof.

$NS(A) \subseteq R^n$

$NS(A)$ is not empty ($\because A\mathbf{0} = \mathbf{0}$)

Let $\mathbf{x}_1, \mathbf{x}_2 \in NS(A)$(i.e., $A\mathbf{x}_1 = \mathbf{0}$ and $A\mathbf{x}_2 = \mathbf{0}$)

Then

1. $A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{0} + \mathbf{0} = \mathbf{0}$
2. $A(c\mathbf{x}_1) = c(A\mathbf{x}_1) = c(\mathbf{0}) = \mathbf{0}$

   Thus, $NS(A)$ is a subspace of R^n

□

Note

The nullspace of A is also called the solution space) of the homogeneous system $A\mathbf{x} = \mathbf{0}$

Example

Finding the solution space (or the nullspace) of a homogeneous system with the coefficient matrix A as follows

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix}$$

Note

The nullspace of A is also called the solution space) of the homogeneous system $A\mathbf{x} = \mathbf{0}$

Example

Finding the solution space (or the nullspace) of a homogeneous system with the coefficient matrix A as follows

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix}$$

Solution

The nullspace of A is the solution space of $A\mathbf{x} = \mathbf{0}$

$$\textit{Augmented matrix} = \begin{bmatrix} 1 & 2 & -2 & 1 & 0 \\ 3 & 6 & -5 & 4 & 0 \\ 1 & 2 & 0 & 3 & 0 \end{bmatrix} \xrightarrow{G.E.J} B = \begin{bmatrix} 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\implies x_1 = -2s - 3t, x_2 = s, x_3 = -t, x_4 = t$$

$$\implies \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s - 3t \\ s \\ -t \\ t \end{bmatrix} = s\mathbf{v}_1 + t\mathbf{v}_2$$

$$\implies NS(A) = \{s\mathbf{v}_1 + t\mathbf{v}_2\}$$

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Rank of a Matrix and Systems of Linear Equations

Theorem ((4.16) The definition of the nullspace)

If A is $m \times n$ matrix, then the row space and the column space of A have the same dimension

$$dim(RS(A)) = dim(CS(A))$$

Rank

The dimension of the row (or column) space of a matrix A is called the rank of A

$$\text{rank}(A) = \dim(RS(A)) = \dim(CS(A))$$

Nullity

The dimension of the nullspace of A is called the nullity of A

$$\text{nullity}(A) = \dim(NS(A))$$

Theorem ((4.17) Dimension of the solution space)

If A is an $m \times n$ matrix, then the dimension of the solution space of $A\mathbf{x} = \mathbf{0}$ (the nullsapce of A) is $n{-}r$. That is,

$$n = rank(A) + nullity(A) = r + (n - r)$$

(n is the number of columns or the number of unknown variables)

Proof.

Since the rank$(A) = r$, the reduced row echelon form of $[A|\mathbf{0}]$ after G.-J.E. should be

$$\begin{bmatrix} 1 & 0 & 0 & c_{11} & c_{12} & \cdots & c_{1,n-r} & 0 \\ 0 & 1 & 0 & c_{21} & c_{22} & \cdots & c_{2,n-r} & 0 \\ & \ddots & & \vdots & \vdots & & \vdots & \\ 0 & 0 & 1 & c_{r1} & c_{r2} & \cdots & c_{r,n-r} & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ & \ddots & & \vdots & \vdots & & \vdots & \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

1. From Thm 4.14, the nonzero row vectors of B, which is the (reduced) row-echelon form of A, forms a basis for $RS(A)$
2. From the definition that $rank(A) = \dim(RS(A)) = r$

Proof.

3. According to the above two facts, the reduced row echelon form should have r nonzero rows like the left matrix

Therefore

$$
\begin{array}{ccccccccccc}
x_1 + & & & c_{11}x_{r+1} & + & \cdots & + & c_{1,n-r}x_n & = & 0 \\
& x_2 + & & c_{21}x_{r+1} & + & \cdots & + & c_{2,n-r}x_n & = & 0 \\
& \ddots & & \vdots & & \vdots & & \vdots & & \vdots \\
& & x_r + & c_{r1}x_{r+1} & + & \cdots & + & c_{r,n-r}x_n & = & 0
\end{array}
$$

Solving for the first r variables in terms of the last $n–r$ parametric variables produces $n–r$ vectors in the basis of the solution space. Consequently, the dimension of the solution space of $A\mathbf{x} = 0$ is $n–r$ because there are $n–r$ free parametric variables.

□

Note

1. rank(A) can be viewed as the number of leading ones (or nonzero rows) in the reduced row-echelon form for solving $A\mathbf{x} = \mathbf{0}$
2. nullity(A) can be viewed as the number of free variables in the reduced row-echelon form for solving $A\mathbf{x} = \mathbf{0}$

Theorem ((4.18) The definition of the nullspace)

If $\mathbf{x}_p$ is a particular solution of the nonhomogeneous system $A\mathbf{x} = \mathbf{b}$, then every solution of this system can be written in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, where $\mathbf{x}_h$ is a solution of the corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$.

Proof.

Let $\mathbf{x}$ be another solution of $A\mathbf{x} = \mathbf{b}$ other than $\mathbf{x}_p$

$\implies A(\mathbf{x} - \mathbf{x}_p) = A\mathbf{x} - A\mathbf{x}_p = \mathbf{b} - \mathbf{b} = \mathbf{0}$

$\implies \mathbf{x} - \mathbf{x}_p$ is a solution of $A\mathbf{x} = \mathbf{0}$

Let $\mathbf{x}_h$ be the solution of $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x}_h = \mathbf{x} - \mathbf{x}_p$

$\implies \mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$

$\square$

Example (Rank and nullity of a matrix)

Let the column vectors of the matrix A be denoted by $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ and $\mathbf{a}_5$

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix}_{4\times 5}$$

1. Find the rank and nullity of A
2. Find a subset of the column vectors of A that forms a basis for the column space of A
3. If possible, write the third column of A as a linear combination of the first two columns

Example (Finding the solution set of a nonhomogeneous system)

Find the set of all solution vectors of the system of linear equation

$$\begin{array}{rcrcrcrcr} x_1 & & & - & 2x_3 & + & x_4 & = & 5 \\ 3x_1 & + & x_2 & - & 5x_3 & & & = & 8 \\ x_1 & + & 2x_2 & & & - & 5x_4 & = & -9 \end{array}$$

Solution

$$
\textit{Augmented matrix} = \begin{bmatrix} 1 & 0 & -2 & 1 & 5 \\ 3 & 1 & -5 & 0 & 8 \\ 1 & 2 & 0 & -5 & -9 \end{bmatrix} \xrightarrow{G.E.J} B = \begin{bmatrix} 1 & 0 & -2 & 1 & 5 \\ 0 & 1 & 1 & -3 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}
$$

$$
\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2s - t + 5 \\ -s + 3t - 7 \\ s + 0t + 0 \\ 0s + t + 0 \end{bmatrix} = s \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 3 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 5 \\ -7 \\ 0 \\ 0 \end{bmatrix}
$$

$$
= \underbrace{s\mathbf{u}_1 + t\mathbf{u}_2}_{\mathbf{x}_h} + \mathbf{x}_p
$$

Rank of a Matrix and Systems of Linear Equations

Note:For any $n \times n$ matrix A with $\det(A) \neq 0$

1. If $\det(A) \neq 0$, $A\mathbf{x} = \mathbf{b}$ is with the unique solution $(\mathbf{x}_p)$ and $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $(\mathbf{x}_h = 0)$. According to Theorem 4.18, the solution of $A\mathbf{x} = \mathbf{b}$ can be written as $mathbfx = \mathbf{x}_p + \mathbf{x}_h$. The result of $\mathbf{x}_h = 0$ implies that there is only one solution for $A\mathbf{x} = \mathbf{b}$ and the solution is $mathbfx = \mathbf{x}_p + \mathbf{0} = \mathbf{x}_p$
2. In this scenario, $\text{nullity}(A) = \dim(NS(A)) = \dim(0) = 0$. Furthermore, according to Theorem 4.17 that $n = rank(A) + \text{nullity}(A)$, we can conclude that $rank(A) = n$.
3. Finally, according to the definition of $rank(A) = \dim(RS(A)) = \dim(CS(A))$, we can further obtain that $\dim(RS(A)) = \dim(CS(A)) = n$, which implies that there are n rows (and n columns) of A which are linearly independent

Rank of a Matrix and Systems of Linear Equations

The relationship between the solutions of $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{x} = \mathbf{0}$ for an $n \times n$ matrix A

<table>
<tr><td></td><td>$\det(A) \neq 0$</td><td colspan="2">$\det(A) = 0$</td></tr>
<tr><td>For $A\mathbf{x} = \mathbf{0}$</td><td>Only the trivial solution $\mathbf{x}_h = \mathbf{0}$</td><td colspan="2">Infinitely many $\mathbf{x}_h$</td></tr>
<tr><td rowspan="2">For $A\mathbf{x} = \mathbf{b}$, the solution is $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$ shown in Thm. 4.18</td><td>$\mathbf{x}_p$ must exist</td><td>$\mathbf{x}_p$ exists</td><td>$\mathbf{x}_p$ does not exist</td></tr>
<tr><td>Only one solution $\mathbf{x} = \mathbf{x}_p + \mathbf{0} = \mathbf{x}_p$</td><td>Infinitely many solutions $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$</td><td>Solution $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$ does not exist</td></tr>
</table>

Theorem ((4.19) Solution of a system of linear equations)

The system of linear equations $A\mathbf{x} = \mathbf{b}$ *is consistent if and only if* $\mathbf{b}$ *is in the column space of* A *(i.e.,* $\mathbf{b}$ *can be expressed as a linear combination of the column vectors of* A*)*

Note:Summary of equivalent conditions for square matrices

If A is an $n \times n$ matrix, then the following conditions are equivalent

1. A is invertible
2. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $n \times 1$ matrix $\mathbf{b}$
3. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
4. A is row-equivalent to I_n
5. $\det(A) \neq 0$
6. $rank(A) = n$
7. There are n row vectors of A which are linearly independent
8. There are n column vectors of A which are linearly independent

Sum of two Subspaces

Sums and Intersections

Let V be a vector space, and let U and W be subspaces of V. Then

1. $U + W = \{\mathbf{u} + \mathbf{w} | \mathbf{u} \in U$ and $\mathbf{w} \in W$ and is called the sum of U and W
2. $U \cap W = \{\mathbf{v} | \mathbf{v} \in U$ and $\mathbf{v} \in W$ and is called the intersection of U and W

Note

Therefore the intersection of two subspaces is all the vectors shared by both. If there are no vectors shared by both subspaces, meaning that $U \cap W = \mathbf{0}$. the sum $U + W$ takes on a special name.

Direct Sum

Let V be a vector space, and suppose U and W are subspaces of V such that $U \cap W = \mathbf{0}$. Then the sum of U and W is called the direct sum and is denoted $U \oplus W$

Note

An interesting result is that both the sum $U + W$ and the intersection $U \cap W$ are subspaces of V

Sum of two Subspaces

Example (Intersection is a Subspace)

Let V be a vector space and suppose U and W are subspaces. Then the intersection $U \cap W$ is a subspace of V.

Solution

By the subspace test, we must show three things:

1. $\mathbf{0} \in U \cap W$
2. *For vectors* $\mathbf{v}_1, \mathbf{v}_2 \in U \cap W$, $\mathbf{v}_1 + \mathbf{v}_2 \in U \cap W$
3. *For scalar* a *and* $\mathbf{v} \in U \cap W$, $a\mathbf{v} \in U \cap W$

We proceed to show each of these three conditions hold.

Example (Intersection is a Subspace)

Let V be a vector space and suppose U and W are subspaces. Then the intersection $U \cap W$ is a subspace of V.

Solution

By the subspace test, we must show three things:

1. $\mathbf{0} \in U \cap W$
2. *For vectors* $\mathbf{v}_1, \mathbf{v}_2 \in U \cap W$, $\mathbf{v}_1 + \mathbf{v}_2 \in U \cap W$
3. *For scalar* a *and* $\mathbf{v} \in U \cap W$, $a\mathbf{v} \in U \cap W$

We proceed to show each of these three conditions hold.

Sum of two Subspaces

Solution

1. *Since U and W are subspaces of V, they each contain* $\mathbf{0}$*. By definition of intersection,* $\mathbf{0} \in U \cap W$
2. *Let* $\mathbf{v}_1, \mathbf{v}_2 \in U \cap W$*, Then in particular,* $\mathbf{v}_1, \mathbf{v}_2 \in U$*. Since U is a subspace, it follows that* $\mathbf{v}_1 + \mathbf{v}_2 \in U$*. The same argument holds for W. Therefore,* $\mathbf{v}_1 + \mathbf{v}_2$*is in both U and W and by definition is also in* $U \cap W$
3. *Let a be a scalar and* $\mathbf{v} \in U \cap W$*. Then in particular,* $\mathbf{v} \in U$*. Since U is a subspace, it follows that* $a\mathbf{v} \in U$*. The same argument holds for W. Therefore,* $a\mathbf{v}$ *is in both U and W and by definition is also in* $U \cap W$

   Hence, $U \cap W$ is a subspace of V.

Note

It can also be shown that the sum $U + W$ is a subspace of V

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Theorem ((4.20) Dimension of Sum)

Let V be a vector space with subspaces U and W. Suppose U and W each have finite dimension. Then $U + W$ also has finite dimension which is given by

$$dim(U + W) = dim(U) + dim(W) - dim(U \cap W)$$

Note

When $U \cap W = \mathbf{0}$, the sum becomes the direct sum and the above equation becomes

$$\dim(U \oplus W) = \dim(U) + \dim(W)$$

Sum of two Subspaces

Example (Dimension of Sum)

For example, let $V = R^2$ and consider the subspaces U and W of V defined by:

$$U = \text{span}([1\,0\,0], [0\,1\,0]) \text{ and } W = \text{span}([0\,1\,0], [1\,1\,0])$$

We can verify that U and W are subspaces of V, and that their dimensions are $\dim(U) = 2$ and $\dim(W) = 2$, respectively.
To find the dimension of $U \cap W$, we need to find the set of vectors that are in both U and W.
Notice that the vector $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$ is in both U and W, so $U \cap W$ is spanned by $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$, which has dimension $\dim(U \cap W) = 1$.

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